

PHYS 141 - Introductory Mechanics

Fall 2024

LECTURES (PAS 201): Mon Fr - 11:00AM-11:50AM

DISCUSSIONS (PAS 170C): Wed - as scheduled in UAccess

LABS (PAS 326): as scheduled in UAccess

This class is scheduled to be taught in the **IN PERSON** modality.

Instructor and Contact Information

Instructor: Dr. Tai Kong

E-mail: tkong@arizona.edu

Office: PAS 575

Office hours: Mon Fr - 1:30-2:30 PM

Course Information

Required Textbooks and Materials:

Sears & Zemansky's University Physics with Modern Physics, 15th Edition by Young and Freedman

Laboratory Manual for PHYS 141

Expected Learning Outcomes:

Recognize the vocabulary and units of mechanics.

Understand the concepts, laws, and principles used in mechanics and generate associations between the concepts and laws.

Apply concepts and laws to both qualitative and quantitative problems.

Work cooperatively within a group on problem solving.

Investigate the behavior of physical systems experimentally.

Course Prerequisites:

MATH 122B, 124, 125, or appropriate Math Placement Level. (Note: If you plan on enrolling in PHYS 142/143 or PHYS 241 the following semester, you should be concurrently enrolled in MATH 129 this semester.)

Course Management

The course is administered via in-person lectures, discussion sessions, and experimental labs. In-person **lectures** will introduce important physics concepts with demonstrations and practice examples. **Discussions** are used to help students develop problem-solving strategies. **Labs** provide an opportunity for hands-on experience with the material covered in lecture.

In-person Lectures: There are two 50-minute lectures per week, consisting of demonstrations, discussions, and guided problem solving. Students will receive a voting card during the first lecture which will be used to respond to in-class clicker questions. Students are expected to bring their voting card to all lectures and participate in answering the clicker questions. There is no penalty for incorrect answers. The instructor will not have spare voting cards, so please be sure not to lose the card you are given. A PDF copy of the voting card will be available on the D2L site, should you need to print another copy.

Discussions: Attendance at the discussion sessions is **required** and you must attend the section in which you are registered in UAccess. The discussion sessions are dedicated to better understanding the lecture material through problem-solving activities. Each session you will be given exercises to work on and the instructor, TAs, and undergraduate preceptors will help you complete the given exercises. The exercises are not intended to be difficult but are meant to help you learn the material. The exercises will be completed in groups of no more than four students and groups will be randomized each session. It is likely that you will be given more work than can be finished during the session, so you are expected to be there the whole time. Answers to the exercises will be available on D2L so you may check your work later. It is in your best interest to try to complete the assigned problems as they will be similar to problems you will see on homework and exams.

Discussion Grading: A quiz will be graded at the end of each discussion session. *Your two lowest discussion score will be dropped at the end of the course.*

Laboratory: All students are required to attend a section of PHYS 141 lab. The grade for the laboratory session is based on student work in lab and on lab reports. The report for a given lab will be due at the beginning of the next lab session. Students **must complete a minimum of 10 labs in order to pass the course**. In other words, if a student completes fewer than 10 labs for the semester, the student will receive a failing grade, regardless of their performance in the other areas of the course.

Attendance and Make-up Policy: It is imperative that you attend all lectures and discussion sessions. There are no make-up lectures or discussion sessions. Therefore, the student is responsible for reviewing and understanding any missed material.

Missed exams may not be excused. If you know you will be absent the day of an exam, please contact the lecturer as soon as possible to schedule a time to take the exam early. If you “suddenly” miss an exam, there must be a valid excuse. Contact the lecturer as soon as possible after a missed exam to work out a time for a potential make-up. Valid excuses include:

Severe illness, Grave family circumstances, Participating in University-sanctioned sporting events with supporting documentation, Job interviews or conferences with supporting documentation, Travel for other classes with supporting documentation. The make-up exam may be different from the original exam.

Note: Personal trips or family vacations are **NOT** valid excuses, even if pre-planned.

Note about over-sleeping an exam: If this or something similar happens to you that would normally **NOT** be excused, immediately notify the instructor via email and present yourself to the examination room and/or instructor's office. Depending on the nature of the situation, the promptness of your email, and the availability of a suitable location, the instructor has the discretion to allow you to take a missed exam, albeit with a large (typically 20%) penalty.

Grading Weights:

Homework:15%

Tutorials:15%

Lab:17%

Midterm Exams (3): 11% each

Cumulative Final Exam: 20%

Grading Scale:

$\geq 87.5\%$	A
$\geq 75\%, <87.5\%$	B
$\geq 62.5\%, <75\%$	C
$\geq 50\%, <62.5\%$	D
$<50\%$	E

This is an absolute grading scale. Thus, 87.5 is guaranteed to be an A, etc. Similarly, 87.4 will be a B. The scale will **NOT** be curved at the end of the course.

For **PHYS 140** students, grading weights will be: homework (21%), Tutorials (21%), midterms (12% each) and final exam (22%). Grading scale is the same as above.

Incomplete grades can only be given if you are passing the course and if only a small portion of the course remains to be completed. Thus, a student who has passed the first two exams and done all the homework but misses the last exam due to a medical emergency can be given an incomplete. Conversely, a student with an average of 35% on the exams cannot be given an incomplete. See: <http://catalog.arizona.edu/policy/grades-and-grading-system#incomplete>.

Examinations

There will be three midterm exams and a cumulative final exam. The tentative dates of the exams are:

Exam 1: Week 5, Sep 23, Monday

Exam 2: Week 9, Oct 21, Monday

Exam 3: Week 14, Nov 25, Monday

Final Exam: Week 17-18

All midterms and the final exam are closed-book. For midterms, each student will be allowed to bring a single 3"×5" note card with information of any kind on both sides of the card. For the final, each student will be allowed a single 8.5" by 11" sheet of paper with handwritten notes on both sides. Use of calculators is permitted during all midterm exams and the final exam; however, cell phone calculators (or internet-capable devices) will not be permitted. All students must bring their own calculator that is not part of a cell phone or internet-capable device.

The midterms will only cover material specified by the instructors. However, note that physics is an integrated science - certain concepts will be introduced early in the course and used repeatedly throughout the semester.

Homework

Homework is the most important part of this class. First, homework is worth 15% of your final grade. Second, it is critical to do the homework, or you will not do well on the exams. You learn physics by working problems, it cannot be memorized.

Homework will be assigned approximately weekly. Therefore, you should be working on the homework every day. **DO NOT GET BEHIND.** Homework will be assigned using Mastering Physics. You can only access Mastering Physics using the course D2L site.

Assignments will be posted ~ 7 days before they are due. *Your two lowest homework assignment score will be dropped at the end of the course.* Adaptive Follow-up assignments are **NOT** for credit.

Since homework will be completed online, it is highly recommended that you maintain a homework notebook where you keep all work from each homework assignment, so that the material may be reviewed later. Working on the homework in groups is expected and encouraged. However, if you are simply completing the homework with someone else's help, you will not do well on the exams.

Homework solutions will be available on D2L after the assignments have been submitted.

Others

Exam Bonus Opportunity: Exams will be graded at 100 points. But the total points on each exam may be higher than 100. Bonus points will also be given to everyone at the end of semester if the anonymous course evaluation survey reaches >80% response rate.

Questions about Exam Grading: It is possible that TAs may make mistakes when grading exams. However, they are just as likely to make a positive mistake as they are a negative mistake. Therefore, these mistakes normally cancel each other out. However, if you would like to request a regrade, you must attach a sticky note to the front of your exam with the problem number that you would like regraded and an explanation of why you think it should be regraded. The ENTIRE

exam will be regraded, which means you could end up with a lower score. Regrade requests must be turned in within one week after an exam is returned.

Available Resources:

The instructor - drop-in hours, scheduled appointments, etc.

Graduate TA drop-in hours

SI Sessions - schedule posted on D2L

The Consultation Room - This room is staffed by a different graduate student each hour and is open approximately 9:00AM-5:00PM Monday through Friday.

The course D2L site which will have homework and exam solutions, etc. A computer lab (PAS 272) which you can use to write lab reports. If you are enrolled in a physics course, your CATCARD should provide you with access to the room.

Code of Academic Integrity

Students are encouraged to share intellectual views and discuss freely the principles and applications of course materials. However, graded work/exercises must be the product of independent effort unless otherwise instructed. Students are expected to adhere to the UA Code of Academic Integrity as described in the UA General Catalog. See: <http://deanofstudents.arizona.edu/academic-integrity/students/academic-integrity>.

Working together on homework is encouraged and recommended. However, the work you turn in must be your own. You may not copy work from other students. Additionally, it is unacceptable to copy solutions from online resources. Cheating in this course will not be tolerated. Any incident of cheating will be reported to the Dean of Students and will become part of your official student record. This means that it could be seen by medical or graduate schools that you may apply to in the future. **If you are caught cheating on an exam**, the minimum penalty you can expect (on top of the incident being reported) is a failing grade in the course.

The university mandates that all students caught cheating must pay for and attend a workshop on academic integrity.

Important Note: Uploading material from this course to a website other than D2L is strictly prohibited and will be considered a violation of the course policy and a violation of the code of academic integrity. Accessing material from this course on a site other than D2L, such as Chegg, Course Hero, etc. during a quiz or exam is a violation of the code of academic integrity. Any students determined to have uploaded or accessed material in an unauthorized manner will automatically fail the class. No exceptions.

Absence and Class Participation Policy

The UA's policy concerning Class Attendance, Participation, and Administrative Drops is available at: <http://catalog.arizona.edu/policy/class-attendance-\ participation-and-administrative-drop>.

The UA policy regarding absences for any sincerely held religious belief, observance, or practice will be accommodated where reasonable, <http://policy.arizona.edu/human-resources/religious-accommodation-policy>.

Absences pre-approved by the UA Dean of Students (or Dean Designee) will be honored. See: <http://deanofstudents.arizona.edu/absences>.

Threatening Behavior Policy

The UA Threatening Behavior by Students Policy prohibits threats of physical harm to any member of the University community, including to oneself. See <http://policy.arizona.edu/education-and-student-affairs/threatening-behavior\ -students>.

UA Nondiscrimination and Anti-harassment Policy

Our classroom is a place where everyone is encouraged to express well-formed opinions and their reasons for those opinions. We also want to create a tolerant and open environment where such opinions can be expressed without resorting to bullying or discrimination of others. As such, the University is committed to creating and maintaining an environment free of discrimination; see <http://policy.arizona.edu/human-resources/nondiscrimination-and-anti-harassment-policy>.

Accessibility and Accommodations

Our goal in the classroom is that learning experiences be as accessible as possible. If you anticipate or experience physical or academic barriers based on disability, please let me know immediately so that we can discuss options. You are also welcome to contact the Disability Resource Center (520/621-3268) to establish reasonable accommodations. For additional information on the Disability Resource Center and reasonable accommodations, please visit <http://drc.arizona.edu>.

Additional Resources for Students

UA academic policies and procedures are available at:

<http://catalog.arizona.edu/policies>

Student Assistance and Advocacy information is available at: <http://deanofstudents.arizona.edu/student-assistance/students/student-assistance>

Subject to Change Statement

Information contained in the course syllabus, other than the grade and absence policy, may be subject to change with advance notice, as deemed appropriate by the instructor.